

Design Document

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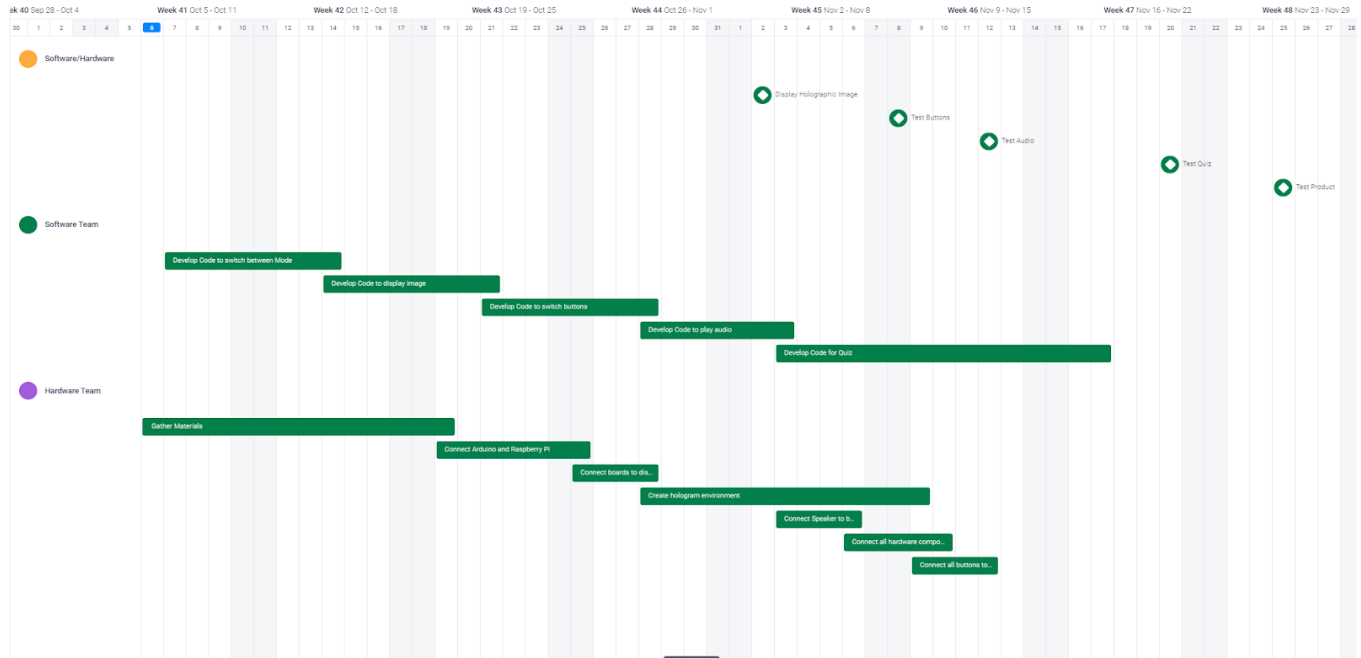
Syed Nayeem, and Khoi Pham

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Gantt Chart



Software Team:

- Oct 7 – 14: develop code to switch between Learning Mode and Quiz Mode
- Oct 14 – 21: develop code to display image
- Oct 21 – 28: develop code to switch between buttons
- Oct 28 – Nov 3: develop code to play audio
- Nov 3 – Nov 17: develop code for Quiz

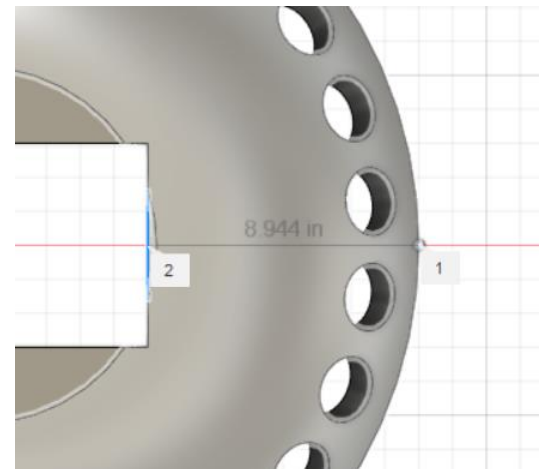
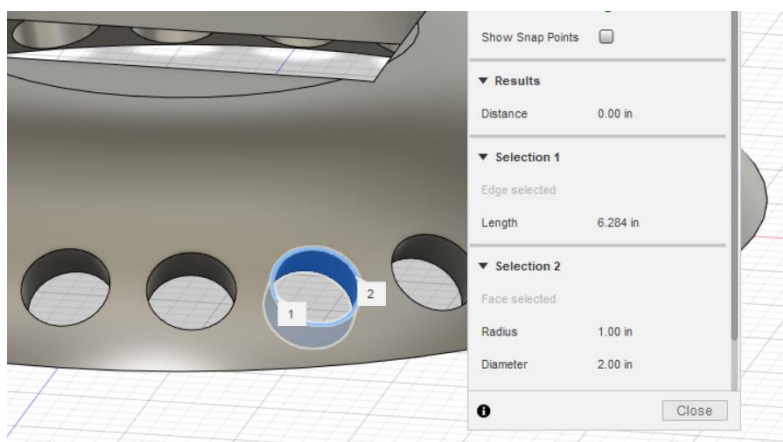
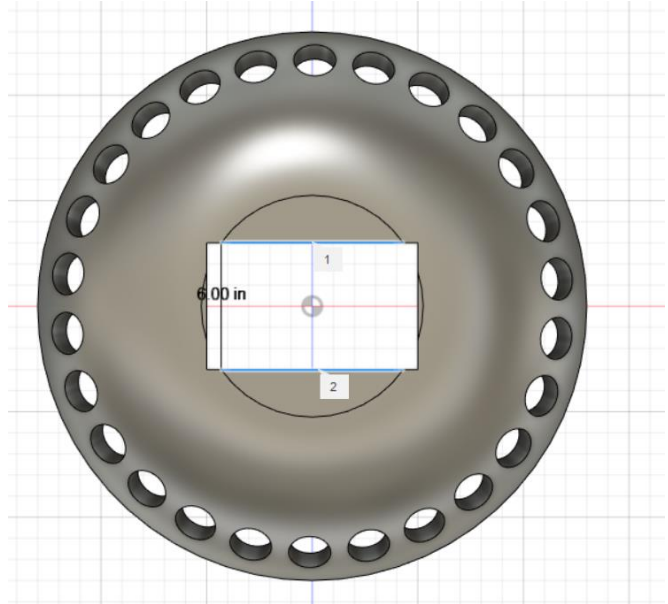
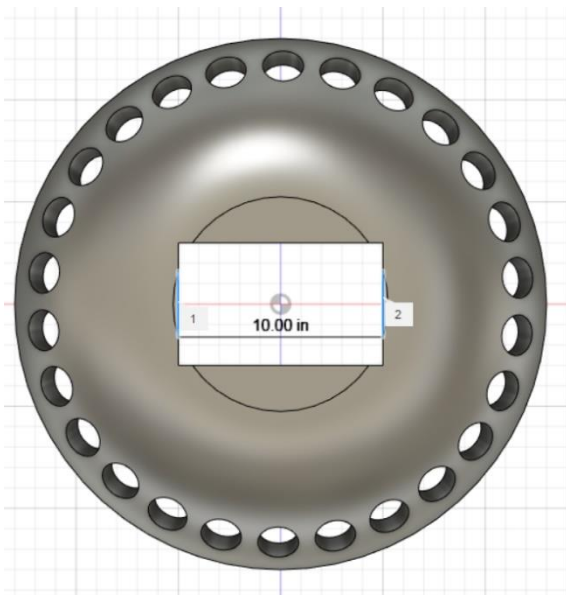
Hardware Team:

- Oct 6 – 19: gather materials
- Oct 19 – 25: connect Arduino to Raspberry Pi
- Oct 25 – 28: connect boards to display
- Oct 28 – Nov 3: create hologram environment
- Nov 3 – Nov 6: connect speakers to board
- Nov 6 – Nov 12: connect all buttons to board
- Nov 12 – Nov 20: connect all hardware components
- Nov 20 – Nov 25: create body of product

Testing Dates:

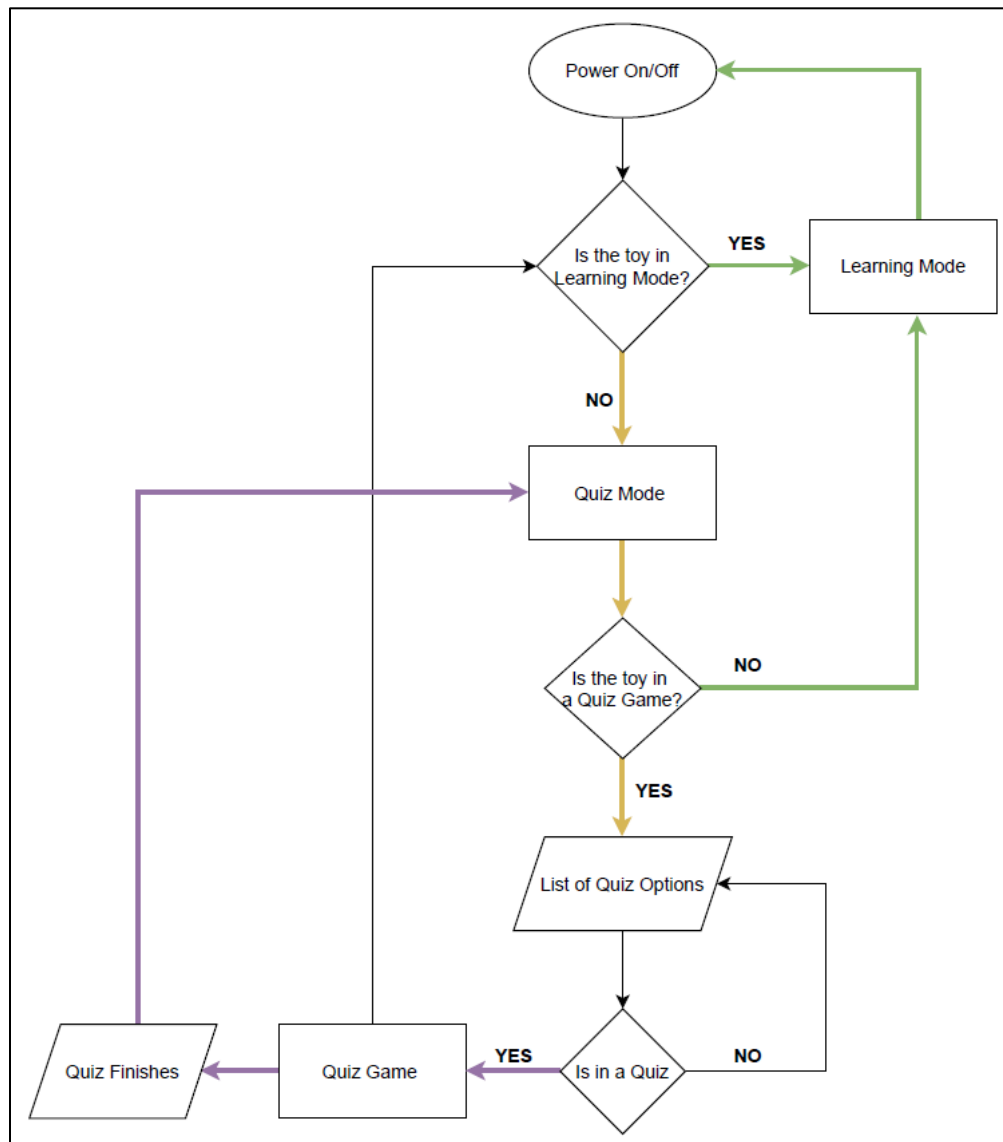
- Nov 3: holographic image
- Nov 8: buttons
- Nov 12: audio
- Nov 20: Quiz Mode
- Nov 27: Product

Model Design

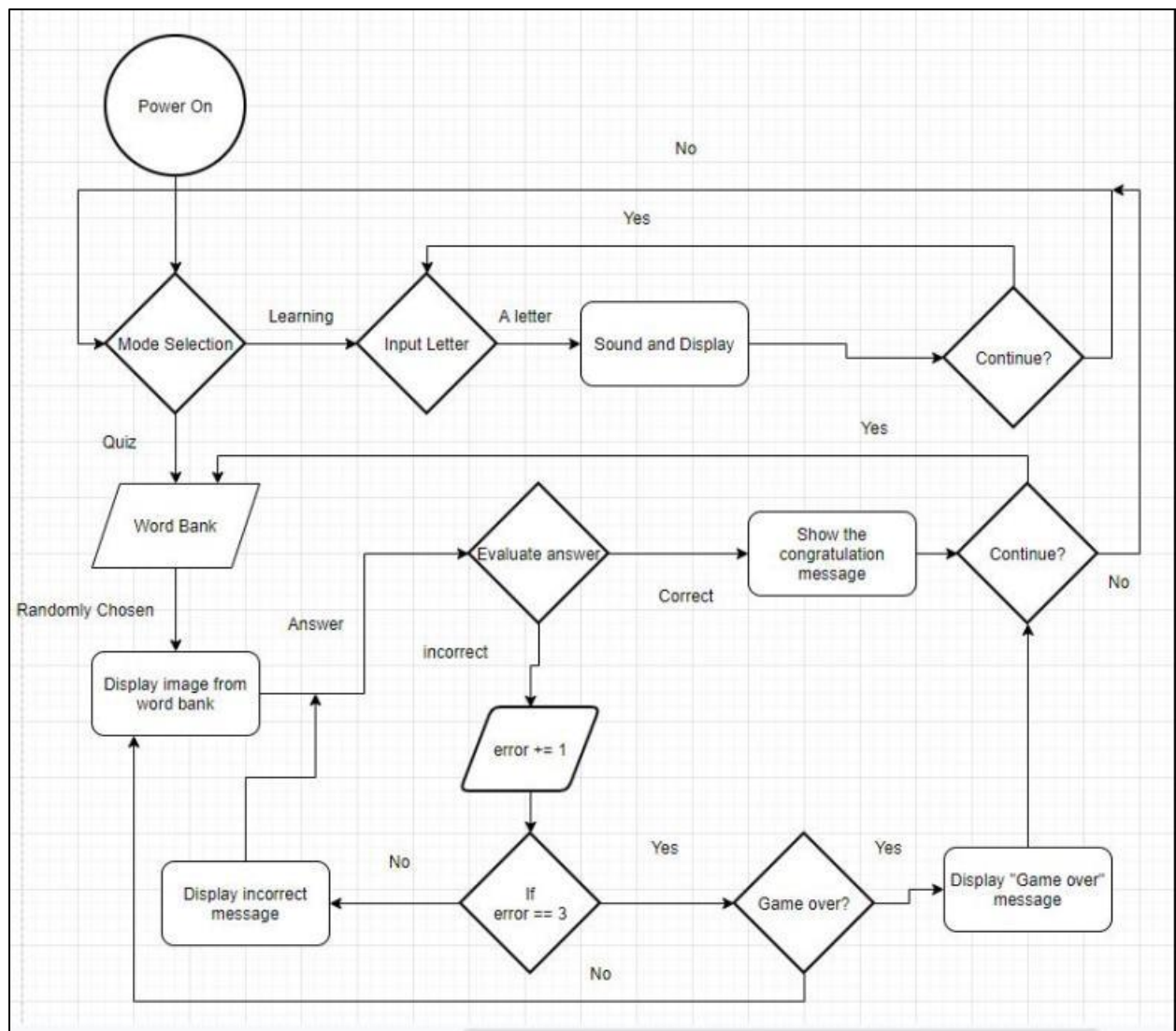


Diagrams

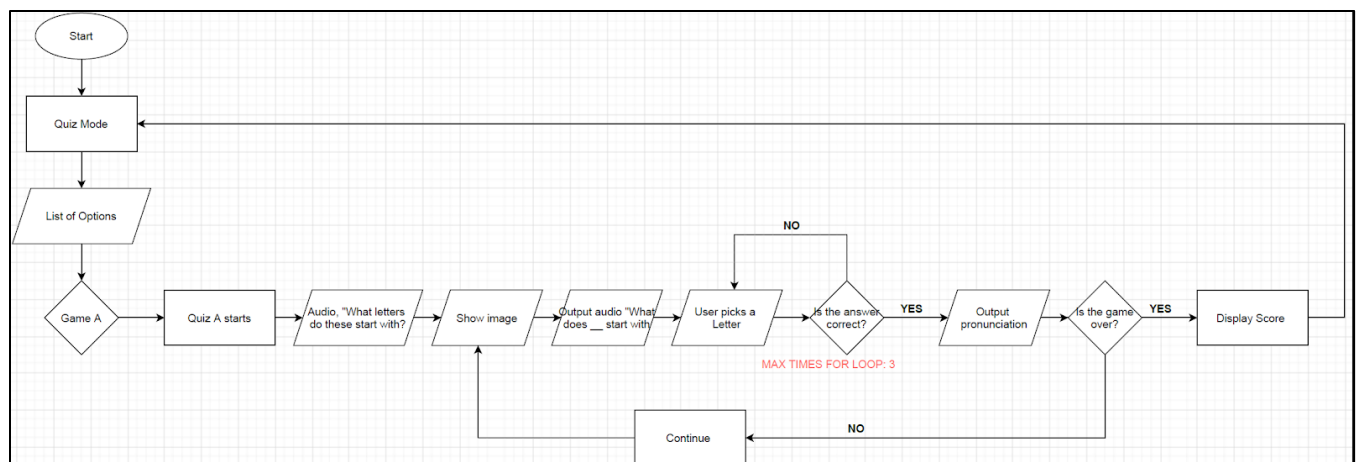
Process Flowchart



UML Diagram



Quiz Mode Flowchart



UML Class Diagrams

Learning Mode

Letter
letter: char imagePath: str audioPath: str
displayLetter(image: str, audio: str)

Quiz Mode

Word
word: str imagePath: str audioPath: str
displayImage(image: str) displayAudio(audio: str)

Quiz
subject: str score: int tries: int = 0 words: array of word objects
playQuiz(words, score, tries) endGameDisplay(score)

Unit Testing

Software Team

- Communication from Arduino
 - Software must interpret pin input data from Arduino correctly
- Audio output functional
- Display output functional
- Mode selection buttons work properly
- Mode switching midgame
- Set up proper databases for letters and quizzes
- Learning Mode
 - Letter objects contain proper sounds and pictures
 - Letter buttons connected to proper letter object in database
- Quiz Mode
 - Correct assessment of inputs
 - Scorekeeping functionality
 - End-game display
 - Quiz selection options at the start of the quiz mode

Hardware Team

- Power On/Power Off button
- Basic input pin functionality (letter buttons)
- Communication to and from Raspberry Pi
- Functioning hologram projection
- Functioning audio
- Functioning switch button to switch between Learning Mode and Quiz Mode